

ZED-RESTO

Report

A Classy Full-Stack Restaurant Web Ecosystem
Scrum Framework | Product Backlog | Sprint Planning

1. Zed-Resto Vision

Zed-Resto is a Modern Full-Stack Restaurant Web Ecosystem designed to deliver a seamless, luxurious digital dining experience — from table reservations and online ordering to a powerful, data-driven administrative back-office. The platform bridges the gap between a classy customer-facing interface and a robust management suite, enabling the restaurant to operate at peak efficiency while maintaining the prestige and warmth of a world-class dining establishment.

Platform Scope & Functionalities

A. Customer Side

- Browse the full menu by category: starters, mains, desserts, and drinks
- Filter dishes by dietary preference (vegan, gluten-free), price range, and guest ratings
- View rich dish detail pages: professional photos, ingredients, allergen information, and chef notes
- Reserve a table online — choose date, time, party size, and special seating preferences
- Order online for home delivery or restaurant pickup
- Guest checkout or full account-based purchase flow with minimal friction
- Apply promotional discount codes at checkout to unlock special offers
- Track order and delivery status in real time
- View full order history with downloadable PDF receipts and invoices
- Write dish reviews and star ratings to guide other diners
- Manage personal profile: photo, contact details, dietary preferences, and loyalty points
- Loyalty rewards program: earn points per visit and redeem for discounts or complimentary items

B. Authentication & Access Management

- Secure login and logout using JWT token-based sessions
- Guest checkout path — no account required for one-time orders
- Role-based access control (RBAC) for five distinct roles:
 - Super Admin, Restaurant Manager, Waiter / Floor Staff, Kitchen Chef, Customer
- Password reset via email verification link
- Profile self-management: update name, photo, email, and password

C. Administrator Side

- Full menu management: add, edit, archive, or delete dishes and categories with photo uploads
- Table and reservation management dashboard: view, approve, modify, or cancel bookings
- Order management center: track dine-in, delivery, and pickup orders with real-time status updates
- User and staff management: create, edit, assign roles, or deactivate accounts
- Sales analytics dashboard: total revenue, top-selling dishes, peak service hours, and trends
- Inventory and stock tracking: low-stock alerts and supplier reorder workflows

- Promotions engine: create percentage or flat-rate discount codes with expiry dates and minimum spend
- Seasonal event scheduler: auto-activate/deactivate themed campaigns (e.g. Ramadan Menu, Valentine)
- Homepage banner and countdown timer management for active promotions
- Customer review moderation: approve, flag, or remove reviews

D. Kitchen / Chef Side (KDS)

- Real-time Kitchen Display System showing incoming order tickets by priority
- Mark individual dishes as 'In Preparation', 'Ready', or temporarily 'Out of Stock'
- View daily mise en place preparation list by shift

E. Delivery Side

- View list of assigned delivery orders with customer address and contact details
 - Update delivery status: Picked Up, On the Way, Delivered
 - Integrated map view with optimized delivery route suggestions
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2. Product Backlog

The product backlog is organized into four Epics covering the complete lifecycle of the restaurant platform, from the guest-facing dining experience through internal administration, security, and marketing.

Epic 1 — Customer Experience

Story ID	Description
US 1.1	As a Diner, I want to browse and filter menu items by category, price, and rating so that I can discover dishes that match my preferences quickly.
US 1.2	As a Diner, I want to place an order as a guest or logged-in user so that I can complete my purchase with minimal effort.
US 1.3	As a Diner, I want to view my order history so that I can track past meals and download invoices.
US 1.4	As a Diner, I want to reserve a table online (date, time, party size) so that I can guarantee my seat without calling the restaurant.
US 1.5	As a Diner, I want to rate and review dishes I have ordered so that I can share my experience with the community.
US 1.6	As a Diner, I want to earn and redeem loyalty points so that I am rewarded for my repeat visits.

Epic 2 — User Identity & Access Management

Story ID	Description
US 2.1	As a User, I want to log in and out securely so that my personal data and order history are protected.
US 2.2	As an Admin, I want to manage user roles and permissions (Admin, Manager, Chef, Waiter, Customer) so that each role has access only to relevant features.
US 2.3	As an Admin, I want to add, update, view, or remove users and customers from the system so that the user base remains accurate and secure.
US 2.4	As a User, I want to manage my own profile including personal information, photo, and dietary preferences so that my experience is personalized.

Epic 3 — Restaurant & Store Management

Story ID	Description
US 3.1	As an Admin, I want to add, edit, and delete menu items with photos, descriptions, and allergen info so that the catalog stays up to date.
US 3.2	As an Admin, I want to change order statuses (Received, In Preparation, Ready, Delivered) so that customers and kitchen staff are always informed.

US 3.3	As an Admin, I want to view a sales dashboard showing total revenue, top dishes, and peak hours so that I can make data-driven decisions.
US 3.4	As a Chef, I want to view and manage my Kitchen Display System (KDS) in real time so that I can prioritize tickets and keep service flowing.
US 3.5	As an Admin, I want to manage table reservations (approve, modify, cancel) so that seating capacity is optimized at all times.
US 3.6	As an Admin, I want to track ingredient inventory and receive low-stock alerts so that I can reorder supplies before running out.

Epic 4 — Promotions & Seasonal Marketing

Story ID	Description
US 4.1	As an Admin, I want to generate unique discount codes (percentage or flat-rate) with expiry dates and minimum spend thresholds so that I can run targeted campaigns.
US 4.2	As a Diner, I want to apply a promo code at checkout so that I can see my total price decrease before confirming payment.
US 4.3	As an Admin, I want to schedule site-wide seasonal events (Ramadan Special, Valentine's Dinner, Summer BBQ) that automatically activate and deactivate at set times.
US 4.4	As a Diner, I want to see event-specific banners and countdown timers on the homepage so that I am aware of current offers and their urgency.

3. Priority Table

User stories are prioritized from 1 (highest) to 3 (lowest) based on business value and technical dependency:

User Story	Priority	User Story	Priority
1.1	1	3.1	1
1.2	1	3.2	1
1.3	1	3.3	2
2.1	1	4.1	3
2.2	2	4.2	3
2.3	2	4.3	3
2.4	3	4.4	3

Legend: Priority 1 = Must Have (green) Priority 2 = Should Have (orange) Priority 3 = Nice to Have (red)

4. Sprint Backlogs

The backlog is delivered across three focused sprints, each building on the foundation of the previous one.

Sprint 1 — The Core Dining Experience

Goal: Deliver a fully functional browsing, ordering, and authentication flow — the restaurant's digital front door.

Story ID	Title	User Story
US 1.1	Browse & Filter Menu	As a Diner, I want to browse and filter menu items by category, price, and rating so that I can find what I need quickly.
US 1.2	Place an Order	As a Diner, I want to place an order as a guest or logged-in user so that I can complete my purchase with minimal effort.
US 1.4	Table Reservation	As a Diner, I want to reserve a table online (date, time, party size) so that I can guarantee my seat without calling.
US 3.1	Menu Management	As an Admin, I want to add, edit, and delete menu items so that the catalog stays up to date.
US 2.1	Secure Login/Logout	As a User, I want to log in and out securely so that my data is protected.

Sprint 2 — Order Management & Administration

Goal: Empower staff with order tracking, kitchen workflows, and user administration tools.

Story ID	Title	User Story
US 3.2	Order Status Updates	As an Admin/Chef, I want to update order statuses so that customers and kitchen staff are always informed.
US 3.4	Kitchen Display (KDS)	As a Chef, I want a real-time kitchen ticket display so that I can prioritize and manage service efficiently.
US 1.3	Order History	As a Diner, I want to view my order history and download invoices so that I can track past purchases.
US 3.5	Reservation Dashboard	As an Admin, I want to approve, modify, or cancel table reservations so that seating is optimized.
US 2.2	Role & Permission Mgmt	As an Admin, I want to manage user roles and privileges based on their responsibilities.

US 2.3	User Management	As an Admin, I want to add, update, view, or delete users and customers from the system.
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Sprint 3 — Business Intelligence & Marketing Growth

Goal: Add analytics, loyalty rewards, inventory control, and a full promotions engine to maximize revenue.

Story ID	Title	User Story
US 3.3	Sales Dashboard	As an Admin, I want a dashboard showing revenue, top dishes, and peak hours so that I can make data-driven decisions.
US 3.6	Inventory Tracking	As an Admin, I want to track ingredient stock and receive low-stock alerts so that I can reorder before running out.
US 1.5	Reviews & Ratings	As a Diner, I want to rate and review dishes so that I can share feedback with the community.
US 1.6	Loyalty Programme	As a Diner, I want to earn and redeem loyalty points so that I am rewarded for my visits.
US 4.1	Discount Code Engine	As an Admin, I want to generate discount codes with constraints so that I can run targeted marketing campaigns.
US 4.2	Apply Promo at Checkout	As a Diner, I want to apply a promo code at checkout so that I can see my total decrease before paying.
US 4.3	Seasonal Event Scheduler	As an Admin, I want to schedule themed events that auto-activate/deactivate at set times.
US 4.4	Event Banners & Timers	As a Diner, I want to see event banners and countdown timers so that I am aware of ongoing deals.
US 2.4	Profile Management	As a User, I want to manage my profile including personal info, photo, and dietary preferences.

5. Recommended Technology Stack

The Zed-Resto platform is built on the MERN stack (MongoDB, Express.js, React, Node.js), extended with carefully selected libraries to cover real-time communication, payments, media storage, and security — everything needed to run a classy, production-grade restaurant web ecosystem.

A. Is MERN Enough?

MERN covers approximately 80% of the platform's needs out of the box. The remaining 20% — real-time Kitchen Display, payment processing, image hosting, and transactional emails — are filled by well-established libraries that integrate seamlessly with the stack. No second backend or alternative framework is required.

B. Full Stack Overview

Layer	Technology	Purpose
Database	MongoDB + Mongoose	Flexible schema for menus, orders, reservations, users, and promotions
Backend API	Node.js + Express.js	RESTful API for all routes: menu, orders, reservations, auth, promotions
Frontend	React + Vite	Fast, component-based UI supporting all five user roles and dashboards
UI Library	Tailwind CSS + Shadcn/ui	Classy, consistent design system — responsive and accessible out of the box
State Management	Redux Toolkit / Zustand	Manage cart, auth session, order state, and notifications globally
Real-Time	Socket.io	Live Kitchen Display (KDS), order status push updates, delivery tracking
Authentication	JWT + bcrypt	Secure login, token-based sessions, role-based access control (5 roles)
Payments	Stripe (or Chargily)	Online checkout, saved payment methods, and automated PDF receipts
Image Storage	Cloudinary	Upload, compress, and CDN-deliver menu photos and user profile images
Email	Nodemailer + SendGrid	Reservation confirmations, order receipts, promo codes, password reset
Maps & Delivery	Leaflet.js (or Google Maps)	Delivery route display and live address autocomplete at checkout
Charts & Analytics	Recharts / Chart.js	Admin sales dashboard: revenue charts, top dishes, peak hour heatmaps

Security	Helmet.js + rate-limit + CORS	Secure HTTP headers, brute-force protection, and cross-origin control
Deployment	Vercel + Railway + MongoDB Atlas	Frontend on Vercel CDN, backend on Railway, database on Atlas cloud cluster

C. Suggested Project Structure

Path	Contents
zed-resto/client/src/pages/customer/	Menu browsing, cart, checkout, reservations, order history
zed-resto/client/src/pages/admin/	Dashboard, menu management, orders, users, promotions
zed-resto/client/src/pages/kitchen/	Real-time KDS ticket display for chef staff
zed-resto/client/src/pages/delivery/	Delivery status board and map view for drivers
zed-resto/client/src/store/	Redux / Zustand slices for cart, auth, and orders
zed-resto/client/src/socket/	Socket.io client connection and event listeners
zed-resto/server/routes/	auth.js, menu.js, orders.js, reservations.js, promotions.js
zed-resto/server/models/	Mongoose schemas: User, Dish, Order, Reservation, Promotion
zed-resto/server/middleware/	JWT auth guard, role checker, rate limiter, error handler
zed-resto/server/socket/	Socket.io server: KDS events, order status broadcasts
zed-resto/server/utils/	Cloudinary helpers, Stripe integration, Nodemailer templates

D. Key Design Decisions

- MongoDB embed vs. populate: Dish snapshots are embedded inside Order documents to preserve pricing history; User references are populated on demand.
- Socket.io rooms: Each order gets its own room so the kitchen, customer, and delivery driver receive targeted real-time updates without broadcasting to all clients.
- Cloudinary transformations: Menu images are auto-resized and served via CDN — never stored in MongoDB, keeping the database lean and fast.
- Stripe webhooks: Payment confirmation is handled server-side via webhook to prevent order fraud from client-side manipulation.
- Chargily alternative: For the Algerian market specifically, Chargily Pay offers a local payment gateway that integrates cleanly with the same Express.js flow.

— End of Zed-Resto Report —